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HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka

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Abstract *Garcinia indica* Choisy Kokum, a tropical fruit and it is a potential source of anthocyanin and the major acid hydroxy citric acid (HCA). HCA is highly present inkokum fruits, which is used as an anti-obesity ingredient in pharmaceutical industries. This study is the aim to evaluate the amount of total proteins, phenols, tannins and other antioxidant derivatives. In addition the targeted to prepare the kokum health juice with different supplements of sugars and pomegranate juice by fermenting with the yeast *Saccharomyces ellipsoideus* isolated from the kokum. The organoleptic test has given a significant result for 20g of sugars and 10ml of pomegranate juice with 7.5 g of kokum rind. Moreover, the anti-obesity citric acid derivative has been estimated from the acetone extract of the kokum rinds and the HPLC report evidence that 45.62 % of HCA and 44.70 % of lactic acids. The significant amount of HCA concentration in aqueous extract being as a potential drug to reduce fat accumulation in the body hence can be used as Anti-obesity agent. As a daily health drink, the fermented juice might be used to avoid a health risks and to avoid a regular medication to reduce obesity.

Key Words *Garcinia Indica*, Hydroxy citric acid, Kokum health juice, anti-obesity, antioxidant.

Introduction

Garcinia indica choicy (kokum) is a small tropical evergreen tree, first discovered in the Western Ghats of India. Kokum fruit is usually cultivated in India with production of 10,200metric tons/haand its very much demand in local as well international. In the year 2004-2005, India exported 77.2 metric tons (Bhat et al. 2005). Estimated demand of kokum fruit in the year 2010-2011 was 894.9 metric tons. *G. indica* is a highly medicinal plant and the bioactive chemical constituents have been isolated from the leaves and fruits including xanthones, benzophenones, flavonoids, biflavonoids and triterpenoids (Rukachaisirikul et al. 2000; Rukachaisirikul et al. 2005, Jamila et al. 2014a; Jamila et al. 2014b; Jamila et al. 2016) with potential medicinal properties like anti-cancer, anti-inflammatory, anti-Alzheimer's, hepato protective and anti-HIV activities (Jamila et al. 2015). Dried fruit rinds of kokum are widely used for gastronomic purposes as a condiment for curries inplace of tamarind or lemon. Conventionally, *G. indica* has been used to treat fever, tumors, ulcers, dysentery, diarrhea and hemorrhoids (Jena et al. 2002; Masullo et al. 2008).

The medicinal property of garcinol has been observed in animal models which provide evidence to explain its potential as a therapeutic agent. According to the drug development directions, garcinol is now in the second phase of preclinical testing, at which theme pharmaco dynamics (PD), toxicology and pharmacokinetic (PK) evaluation should be accomplished before the drug could be processed (Liu et al., 2015). At present, its

fruit juice is widely available as anti-obesity products with regard to their safety/toxicity; kokum has been stated that which was very safe for human consumption.

Earlier medicinal compounds, studies have been shown that *G. indica* fruit rind hasthe highest concentrations of anthocyanins including cyanidin-3-O-sambubioside andcyanidin-3-O-glucoside (Jamila et al., 2016). Anthocyanin has potential biological activities like anti-inflammatory, anti-cancer, anti-oxidant, anti-diabetic, anti-obesity and anti-apoptotic activities. However, the identification and purification of (-)-hydroxycitric acid [(-) -HCA] from *Garcinia indica* and its biological activities have fascinated the devotion of biologists and healthexperts. The biological effects of (-)-HCA have been studied expansively for its unique regulatory effect on lipo genesis, fatty acid synthesis, weight loss and appetite. The derivatives of (-)-HCA have been incorporated into a wide range of therapeutic combination with other ingredients to enhance weight loss, reversing lipid abnormalities and cardio protection.

Pomegranate, a phyto nutrient rich fruit, falls into the super fruit Category. The juice and functional beverages of the fruit deliver nutritional benefits like vitamins, fatty acids, minerals and anti-oxidants into the human diets. Among processed juices, pomegranate is considered to have higher market value (Rymon, 2011).Fruit juices are normally fermented to increase their shelf-life and aid distribution before consumption. In fact, the applied heat might impact on undesirable biochemical and nutritional changes (Vegara et al., 2013). The most attractive method to process the juices is the lactic acid-fermented pomegranate juice (PJ) (Filannino et al., 2013). Lactic acid fermentation method currently used as the preservative method or the manufactured products. In the fermentation processes the microbes can utilize pomegranate juice and kokum rind juice as a nutrient and produce the biochemical compounds as secondary metabolites. Fermented products have been studied for its therapeutic effect and they are highly active on cytotoxicity, anti-oxidant and anti-inflammatory activities (Liu et al., 2015). Although, most of the gaseous compounds are naturally accumulates in the plants, they might be processed as secondary metabolites by microbes (Sarry and Günata, 2004; Di Cagno et al., 2009) in an anaerobic fermentation.

Obesity has reached as an epidemic proportion resulting from the imbalance between intake and energy disbursement, as a significance of a sedentary lifestyle and poor eating habits (Sethi 2011). This epidemic is a significant global health problem that affects adults and children worldwide. Obesity is categorized as a chronic disease affected by excessive deposition of body fat. Despite researchers' efforts to understand the mechanisms that lead to overconsumption of food and overweight (Fassina et al., 2015).Besides, highly pigmented fruits as pomegranate often have sour, tart and mordant components that may make the products undesirable, especially when processed. Despite of importance for PJ and the key role of volatile flavor and medicinal compounds on the consumer acceptance, aroma deviations, potential flavors development and flavor preferences among consumers have not been fully studied and established (Mayuoni-Kirshinbaum and Porat, 2014).

Surgery and synthetic drugs are usedas a therapeutic treatment for obesity, which can result in high costs and serious complications. Natural agents, herbal products are an

alternative approach to manage weight, in attacking obesity resulting in a safer and highly effective treatment without risks. Thus, the organic compounds a new potential medicine or alternative therapy for the treatment of obesity and frequently been used to manage weight.

Production of natural juices with the medicinally important organic compounds would improve the health and wealth in day-to-day life. Despite of the consumers are looking for more natural foods with the more nutritional and therapeutic values to avoid a daily medication. Moreover, the microorganism has a potential to modify the substrate to enhance their nutritional and functional characteristics, which could change the natural compounds into a more active products. This study aim to quantify the bioactive compounds such as ascorbic acid, glutathione, tannins, phenols and proteins in the *Garcinia indica* choicy dried rind methanol extract and fermented kokum rind juice with the highly pigmented fruit pomegranate juice. Also we have evaluated the fermented juice for its safety/toxicity acceptance by organoleptic test to evidence the fermented juice for oral consumption. In addition estimate the hydroxycitric acid in the fermented *G. indica* fruit, rind juice for its anti-obesity potential.

Materials and methods

Sample collection and kokum rind preparation

Garcinia indica choicy Kokum fruit samples were collected different places of transitional zones of Karnataka, India, namely Udipi, Mangalore, Kasingud, Goa and Sirsi. The rinds of kokum fruits were separated and dried by sunlight for 15 days. After the kokum rinds dried which was used for the biochemical estimation and characterization.

Total protein estimation

100 mg of *Garcinia indica* (kokum) dried rind was extracted with 0.1M Phosphate buffer (pH 7.0) for 1 hour on a magnetic stirrer. The extract was centrifuged at 10,000rpm for 20 minutes and the supernatant was used for the estimation of total soluble protein by Lowry's method (Lowry et al., 1951) with BSA as standard. The OD is taken at 660nm against the blank. A standard graph is plotted and the unknown concentration of protein is determined.

Total phenol estimation

100 mg oven dried sample was extracted with 10mL of 80% ethanol for one hour a room temperature. The extract was centrifuged at 6000 rpm for 15 min. The supernatant was evaporated to dryness a water bath and the residue were dissolved in 5ml of distilled water. The alcohol free extract was used for the estimation of phenols (Malick et al., 1980). A different aliquot of catechol ranging from 0.2 to 1.0 ml was taken in different test tubes and 1.0 ml of test sample was taken for phenol estimation. The volume is made up to 3 ml with distilled water and 0.5ml of FCR is added and incubated for 3 minutes. 2 ml of 20% Sodium carbonate is added to all the test tubs and kept in boiling water bath for 1minute and cooled under tap water. 3ml of distilled water is added and the OD is taken at 650nm in a spectrophotometer. A standard graph is plotted and the unknown concentration of phenol is determined.

Total tannins estimation

100 mg of air dried *Garcinia indica* (kokum) rind was extracted with 5ml of methanol for 24 hours at room temperature

with stirring. The extraction was centrifuged at 5000 rpm for 10 minutes. The supernatant was used for the estimation of total tannins. 1 ml of aliquot sample was added with 5ml of vanillin-HCL reagent and mixed thoroughly then incubated the tubes for 20 minutes at room temperature. The absorbance was recorded at 500 nm against the blank reagent by using UV-spectrophotometer. The standard graph was constructed with standard catechin the concentration of 20 µg, 40 µg, 60 µg, 80 µg and 100 µg. The graph was plotted for unknown samples and calculated the tannins concentration expressed per gram of oven dried samples.

Estimation of ascorbic acid by iodine titration method

5g of dried *Garcinia indica* (kokum) rind powder was dissolved in 50 ml of distilled water. Added 25ml of vitamin C (250 g of vitamin C in 100 ml of distilled water and dilute to 200 ml with distilled water) standard solution/Sample in to 125 ml of conical flask. Then 10 drops of 1% Starch were added. Titrate the samples with iodine solution (5g of Iodine and 0.268 g of potassium iodide dissolved in 200 ml of distilled water and add 30 ml of 3M sulfuric acid and diluted to 500 ml with distilled water) against the sample/standard to reach the end point as dark blue.

Estimation of glutathione by Ellmann's (1959) Method

3g of sun dried *Garcinia indica* (kokum) rind was suspended in 25 ml of methanol and mixed well and kept it for 12 hours at 3°C. Then homogenized it and centrifuged at 15000 rpm for 20 minutes. The supernatant was stored at -20ºC till use. Standard glutathione (0.5 mg/ml) was prepared with 0.2, 0.4, 0.6, 0.8 and 1.0 ml volumes in five test tubes respectively. Then the tubes were made up to 2.0 ml with distilled water. 2.0 ml of distilled water was prepared as a blank and 2.0 ml of methanol extract was also added for unknown test samples. Then 3 ml of precipitating solution was added to each test tube and mixed solutions were kept for 5 minutes at room temperature. Then the solutions were filtered with normal filter paper and 2 ml of filtered solution was taken for further process. Then 1 ml of DTNB solution (40 mg of DTNB was dissolved in 1% of Sodium citrate) was added to the test solution and colorimetric absorption was recorded at 412 nm. The graph was plotted and calculated the glutathione concentration expressed per gram of oven dried samples.

Estimation of hydroxy citric acid (HCA)

HCA was extracted from the dried fruit rind with acetone. Acetone extract was concentrated and diluted with water and lyophilized. It was used for HCA estimation. 50 g of air dried rind autoclaved with 150 ml of water at 115°C for 15 minutes. The cooled extract (25-30°C) was decanted in cheese cloth and Whatmann No.1 filter paper 150 ml of filtered extract was concentrate to 25 ml on water both and treat with 100 ml of 60% ethanol. The reacting precipitate was centrifuged to remove dry mate. Acid filtrate is neutralized by adding of 40% KOH with careful stirring. Heavy oily liquid was formed and allowed to settle down and the supernatant was discarded. The oily liquid was washed with 25 ml of 60% ethanol for 5-6 times. Then wash the absolute ethanol for 2 times. The samples should be left for 60-90 minutes each time. 150 ml of absolute alcohol was added and allowed to stand the solution. Ethanol was decanted and yellow (very hydroscopic) semi

solid was obtained and dried in vacuum at 80ºC to remove ethanol. Dried powder was weighed and calculated the HCA content for 50g.

Fermentation study for the preparation of kokum health drink

The kokum fruit rinds were sun dried for 10-15 days and soaked in different concentration as 25g/lit, 50g/ lit and 75g/ lit for overnight. Then samples were homogenized and filtered by using muslin cloth. YEPD media (Yeast extract -10g/lit; Peptone -20g/lit; Dextrose -20g/lit; pH- 5.4) was used for isolating yeast culture from the kokum rind samples by serial dilution. The broth cultures were incubated at 28°C ± 2 for 24 hrs. The filtrates were taken in different flasks in different concentration as given in the table. 1.

All the treatment combinations were treated with 200ppm potassium meta bisulphate and kept for 1-2 hours. Initially all the solutions changed its colour to yellowish pink and waited till it turns to normal colour of the juice. Then 5ml of yeast (Saccharomyces ellipsoideus) culture was added to all the flasks and kept them for fermentation 12-15 days. After the fermentation process the contents were filtered using normal filter paper and were pasteurized at 62ºC for 15-30 minutes. The pasteurized kokum juice is used for further studies.

Organoleptic evaluation for fermented kokum juice (*Garcinia indica* Choisy)

Organoleptic test was carried out to determine the acceptability and palatability of the fermented fruit juice by the best graded system (Sunilkumar et al., 2003). The organoleptic test was evaluated by M.Sc Biotechnology students, University of Agricultural science, Bangalore namely Ms. Sowmya NS, Ms. Mingkee A, Ms. Sowjanya M S, Ms. Manasa D S, Mr. Bharath prasad C D, Ms. Namrata S. Pawar, Ms. Humera jabeen, Ms. Swetha, Ms. Deshmukh snehal, Ms. Rini pakarla and Mr. Lingaraj Dalawai. Standard grade scores and their respective remarks for fermented juice were recorded.

High performance liquid Chromatography analysis

HPLC analysis was performed for aqueous extract of kokum rind.20 µl of samples were injected in C18 Zorbax columns of HPLC chromatogram (Thermo Finnigan, Europe) with SRVYR-LPUMP, Spectrum SN4000 and Wavelength of 215nm at ambient temperature. Finally elated with Ethanol and Water in the ratio 65:35 at a low rate of 1 ml/ min.

Statistical analysis

Each experiment was performed in triplicate and the results are expressed as the mean ± standard deviation (SD). The one-way analysis of variance (ANOVA) was used for comparing the data between the samples as well as the test groups. P-values < 0.05 were considered significant.

Result

Amount of total protein from kokum rinds

As the Figure 1 shows the result of biochemical test for total proteins varies from 1.96 to 1.60 mg/g of the samples analyzed all the samples which were analyzed for total protein were not differing much concentration, since all are on par with each other. Thus the present result depicts uniformity in protein distribution in all the regions of the Western Ghats of Karnataka.

Amount of total phenolic from kokum rinds

As the Figure 1 shown the samples collected from the Sirsi as given the high content for phenol (0.88mg/g) as comparatively as to other samples. The lower content was reported further samples of Udipi samples (0.66mg/g). All the other samples were found in the moderate range of 0.81 mg/g in the region of Goa, Kasingud and Mangalore. Medicinal value always records for higher content of phenol, the significance level was reported from the analysis of a sample collected from sample Sirsi (0.88 mg/g).

Amount of total tannins from kokum rinds

Phenol content in the kokum rind was analysed and is present in the figure 1. The Figure shows that the samples collected from the Udupi as given the high content for tannins (1.55 mg/g) as compare to other samples. The lower content was reported from Mangalore samples (0.20 mg/g). All the other regions Goa, Kasingud and Sirsi samples were shown moderate range from 0.50 to 1.50 mg/g.

Amount of total glutathione from kokum rinds

Glutathione content in the kokum rinds was analysed and is presented in the figure 2. The table shows that the samples collected from the Udipi as given the high content for Glutathione (1.68 mg/g) as compare to other samples. The lower content was reported Mangalore samples (1.41 mg/g). All the other samples were in the region of Goa, Sirsi and Kasingud (1.50 to 1.55 mg/g).

Amount of total ascorbic acid from the fermented kokum juice and Methanol extracted kokum rinds

Ascorbic acid content in the fermented kokum juice and methanolic extract of kokum rind were analysed and the result shows that ascorbic acid concentration is significantly increased in the fermented kokum juice supplemented with 10 ml of pomegranate juice to compare with methanolic extract of kokum rind in the range from 104.40 to 304.00 mg/lit. The treatment T8 shows high content for Ascorbic acid (304.00 mg/lit) (Fig. 2).

Development of fermented health juice from Kokum rinds by using isolated yeast culture *Saccharomyces ellipsoideus*

Isolated cultures were growing on the YEPD media and sent for identification to confirm as yeast cultures Saccharomyces ellipsoideus in the MTCC, Chandigarh. This culture, which was isolated from the Kokum performs well for the fermentation process. Kokum juice was fermented in various concentrations of rinds (2.50g, 5.00g and 7.50g/100 ml water), sugar (10, 20 and 30 g) and 10 ml of pomegranate juice then fermented for 15 days at room temperature and the fermented juice was subjected to organoleptic test.

Organoleptic evolution for fermented kokum juice (*Garcinia indica*)

The test result shows that treatments T7 and T8 as given high score 25 and 28 respectively as compared to other treatment. The treatments were graded based on the organoleptic score by comparing with the standard grade scoring system. The treatment T7 and T8 come under the grade first (Fig. 6). So the sample T8 is recommended for the further study to go for fermentation in large scale. This large scale fermented juice may be used for medical value of health drink of kokum.

Hydroxy citric acid (HCA) content in kokum rinds

Hydroxy citric acid content in the kokum rinds was analysed and is presented in the figure 3. The table shows that the samples collected from the Udipi as given the high content for HCA (0.41 mg/g) as compare to other samples. The lower content was reported Mangalore samples (0.32 mg/g). All the other samples were in the region of Goa, Sirsi and Kasingud (0.34 to 0.37 mg/g).

HPLC analysis for the kokm rinds (*Garcinia indica*)

HPLC chromatogram (Fig. 4 & 5) was revealed that a significant content of Hydroxycitric acid (45.62%) and Lactone (44.70%) was recorded and 8.98% citric acid also reported from the aqueous extracts of rinds. HCA (Hydroxy citric acid) is of significant medical importance, being the compound helps to reduce fat accumulation in the body hence can be used as Anti-obesity agent.

Discussion

The genus Garcinia is a group of well-known fruit trees in Madagascar and Southeast Asia. The fruit of many species are palatableand is used as a substitute for tamarinds in food. Certain species have a potential effect on different types of medicinal values like antimicrobial and pharmacological activities (Valdir et al., 2000).*Garcinia indica* is one of these, and its fruit is rich in garcinol. The derivative of garcinol has various therapeutic imports like anti-inflammatory, anti-cancer, anti-Alzheimer, anti-diabetic and anti-obesity (Chen et al., 2008). Moreover, the fruit hulls of *Garcinia indica* also contain several types of abundant such as xanthones, benzophenones, flavonoids, biflavonoids and triterpenoids (Jamila et al. 2016). Kokum products as atherapeutic drugs produces decrement in LDL, TG and TC serum levels. In an inhibiting effect of hydroxy citric acid on ATP citratelyase, an enzyme which catalyses extramitochondrial cleavage of citrate to oxaloacetate and acetyl CoA. These findings suggest that Garciniahas hypolipidemic action causing significant hypocholesterolemic and hypotriglycerdemic effects (Amin et al., 2011).

The total phenolics (TPH) content (Fig.1) had been reported per dry matter of the fruit rinds. The rind and skin inhabit over 70% of the weight of kokum fruit. The yields of total protein and phenols have ranged from 15.0% to 20.5% and 5.8% to 7.9% of fruit part weight respectively. The extraction yield from rind is greater than did skin. These values are very much higher than the previous reports (Okonogi et al., 2007). Catechol is a common standard used for the quantification of phenolics extracted from plant materials. The kokum rind and skin contains over 25-fold more total phenols than edible aril (Pothitirat et al., 2009). Numerous factors are influences in the total phenol yield in plant materials such as regional differences (Häkkinen and Törrönen, 2000), the degree of ripeness (Mosel & Hermann, 1974), harvest conditions, time, temperature as well as the methodical procedure used for the extraction and quantification of the phenols and proteins. Facts of attractions of plant phenolic and protein mixtures are significant as utilized as food antioxidants and the interactions of phenolics on their antioxidant potential (Naczk et al., 2011).

The total content of tannins (Fig. 1)in kokum was determined by the vanillin and proanthocyanidin assays. The vanillin assay is frequently used for estimation of total tannins due to its specificity for flavanols and dihydro

chalcones (Sarkar & Howarth, 1976). Plant phenolics extracts are combinations of polymeric derivatives of tannin compounds that differ in their sensitivities, structures and their functions in different therapeutic uses. Huang et al. (2007) also reported that mangosteen trees are an excellent source for type B proanthocyanidins. Although, tannins of Garcinia have an antibacterial effect by precipitating theirproteins making them unavailable their nutritional proteins (Fig. 1). Kokum extract recorded the smallest MIC and MBC has been studied on *Salmonella typhimurium* and *Salmonella serover* (Ngoupayo et al., 2015).

Eventhough the increase of antioxidant ability may be related to R-mangostin, other antioxidants. Vitamins C and E are well known antioxidants contained in this xanthone-rich product. However, significant increases of vitamins C and glutathione induce more than double the antioxidant values (Kondo et al., 2009). The increase of vitamins C and E in plasma would not be sufficient to give such a large impact on chronic diseases, and there is no potentiating observed on the antioxidant effects by vitamins C and glutathione due to the high base concentrations (Kondo et al., 2009). Although the microbial fermentation could increase the potential substrate/co adjuvant production to increase the recommended daily values (Chaoqun et al., 2015).

Phenolic acid metabolism contributes to the stress response in the anaerobic fermentation process and produces the secondary metabolites. Moreover, the catabolism of phenolic acids by forcing heater fermentative species is linked to the reoxidation of the reduced cofactors and thus affords an energetic advantage. Phenolics metabolism may thus confer a selective advantage for yeast in vegetable and fruit fermentation (Huang et al., 2007). Phenol byproducts may also utilize biological activities. Therefore, bioconversion pathways could be synthesis as derivatives of pharmaceutical important products (Filannino et al., 2015).

Hydroxycitric acid (HCA) (Fig. 3) is a derivative of citric acid that is found in kokum plant and which has highly anti-obesity effect. Reports have shown that consumption of hydroxy citric acid decreases appetite, halts fat lipogenesis, synthesis, decreases food consumption and reduces body weight (Preuss et al., 2004). HCA is nontoxic as organoleptic study has shown that by oral route it did not cause any behavioral and systematic toxicity even at high dose of 5 g/kg body weight (Jena et al., 2002). HCA is also reported that avoids toxic effects like nervousness, fast heart rate, high BP, or insomnia signs in humans, thus which has no toxic or humans in nature (Preuss et al., 2004).

Conclusion

Garcinia indica choicy Kokumown various health benefits and that the phytochemicals garcinol, anthocyaninsand (−) –hydroxycitric acid are useful in various medical conditions. Sirsi samples having high content of phenol and HCA this sample could use for future studies against the various diseases. HPLC result also revealed that a significant amount of HCA has reported from the aqueous extract, thus the medicinal importance being the compound helps to reduce fat accumulation in the body hence can be used as Anti-obesity agent. The combination of kokum rind 7.5 g and sugar 20 g fermented with *Saccharomyces ellipsoideus* and fortified with 10 ml of pomegranate juice

could be recommended for the large scale production of "KOKUM HEALTH JUICE".

Acknowledgments

The authors would like to thank the Department of Biotechnology, Bishop Heber College, Tiruchirapalli, Tamil Nadu, India for their encouragement and help to provide very useful information and facilities.

Table 1: The combination of treatments to prepare kokum health juice

S. No	Treatments	Rind (g)	Sugar (g)	P.G.J (ml)
1.	T_1	2.5	10	10
2.	T_2	2.5	20	10
3.	T_3	2.5	30	10
4.	T_4	5.0	10	10
5.	T_5	5.0	20	10
6.	T_6	5.0	30	10
7.	T_7	7.5	10	10
8.	T_8	7.5	20	10
9.	T_9	7.5	30	10
10.	T_{10}	5.0	Not added	10
11.	T_{11}	5.0	Not added	Not added

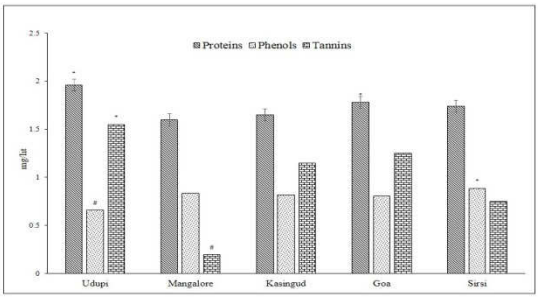


Fig. 1: The biochemical analysis for total protein, total phenol and total tannin contents in kokum rind from different regions of Karnataka

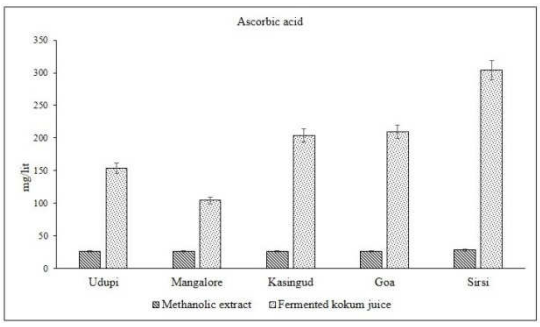


Fig. 2: The biochemical analysis for Ascorbic acid from methanol extract of kokum and the fermented kokum health juice and glutathione contents in kokum rind from different regions of Karnataka

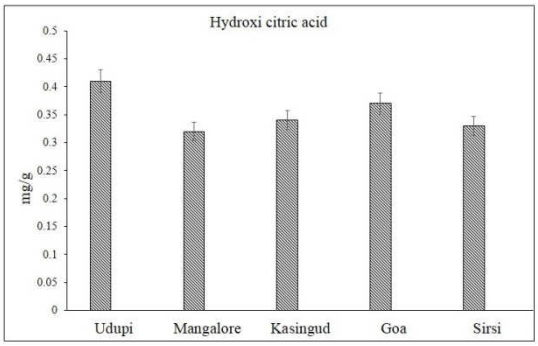


Fig. 3: The biochemical estimation of hydroxycitric acid from acetone extract of kokum rind from different regions of Karnataka

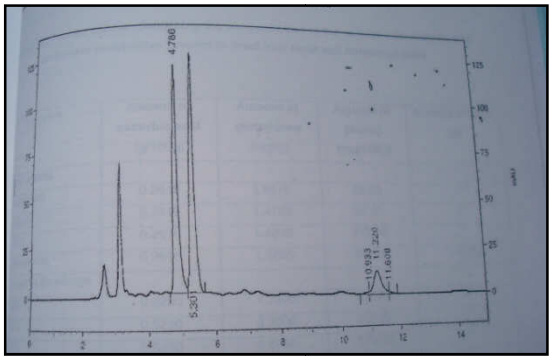
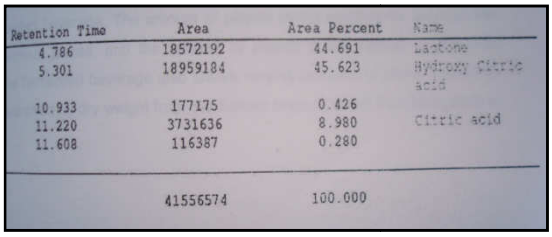


Fig. 4: The HPLC chromatogram of hydroxycitric acid from aqueous extract of kokum rind



Retention Time	Area	Area Percent	Name
4.786	18572192	44.691	Lactone
5.301	18959184	45.623	Hydroxy Citric acid
10.933	177175	0.426	
11.220	3731636	8.980	Citric acid
11.608	116387	0.280	
	41556574	100.000	

Fig. 5: The HPLC chromatogram result of hydroxy citric acid from aqueous extract of kokum rind

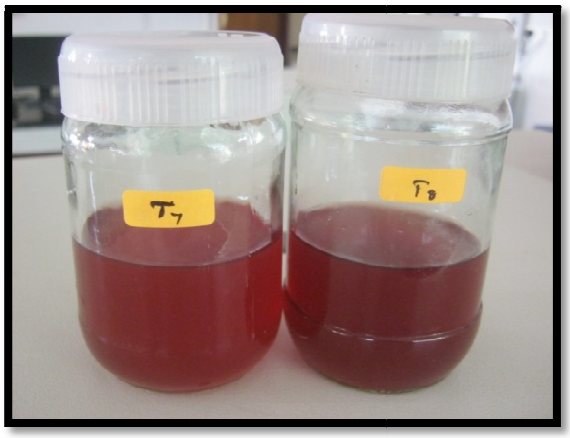


Figure 6: First graded fermented Kokum (*Garcinia indica*) health juice samples by organoleptic test

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